

Forecasting LA Home Values

Mortgage rates more than doubled after 2021. Did LA home values get harder to predict?

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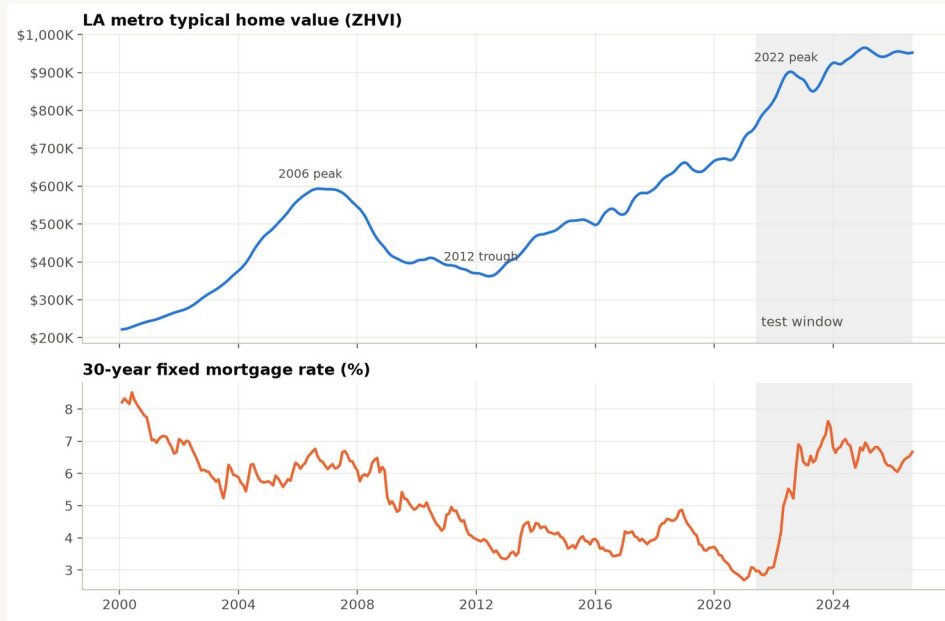
Zillow Home Value Index, LA metro, Jan 2000 to Aug 2026, plus 499 neighborhoods and the 30-year mortgage rate



Solid: actual · Dashed: our forecast to Aug 2027

THE QUESTION

LA values grew 5.6% a year for 26 years, then stalled when rates hit 7%



\$952,601

typical LA-metro home, August 2026

-1.3%

vs the January 2025 peak of \$965,634

2.7% → 7.6%

30-year mortgage rate, 2021 low to 2023 high

We tested 14 models two ways, so no model could win by luck

DATA

Zillow ZHVI, LA metro

320 months, Jan 2000 to Aug 2026

499 LA neighborhoods

Zillow neighborhood index

Mortgage rate and payroll jobs

FRED and BLS, LA and Orange County

14 MODELS + 3 LLMS

Baseline and classical

Naive drift · ARIMA · Holt · Prophet ×2

ML, deep learning, AI

XGBoost · LightGBM · LSTM · GRU ·
Chronos

Outside information

ARIMAX ×2 · VAR · factor VAR

Plus 3 chat LLMS

Claude · GPT · Gemini

2 TESTS

Blind 80/20

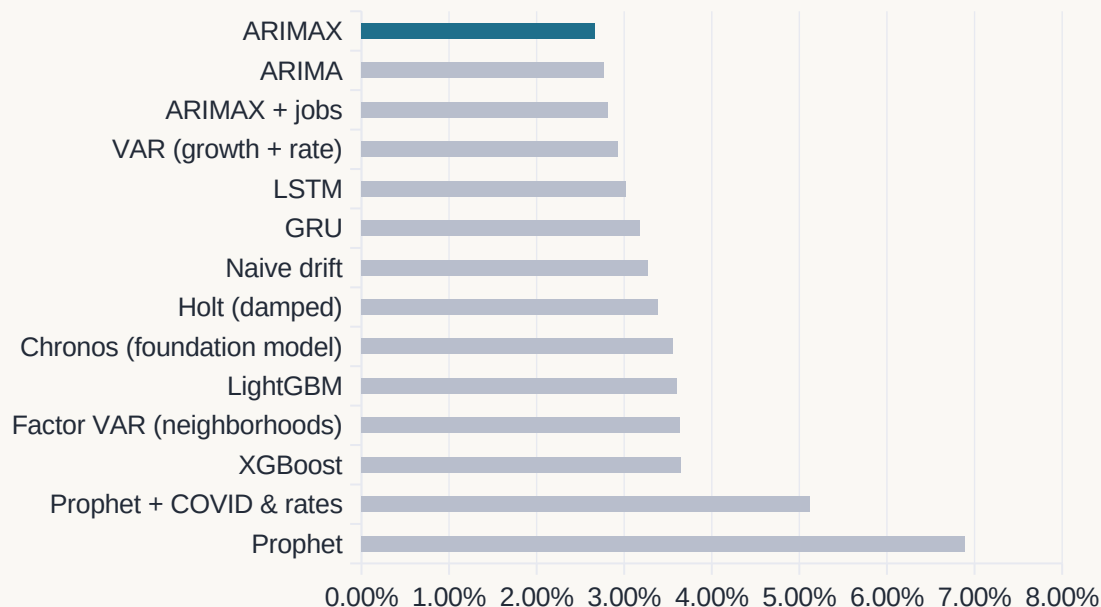
Train to Apr 2021, forecast 64 months in
one shot

Rolling origin

Refit every 3 months, forecast 12 months
ahead, 18 times

WHY NOT 1-MONTH ERRORS Zillow smooths this index, so every decent model is about 0.1% off one month ahead. We judge models at 6 to 12 months instead.

In the fair test, ARIMA-family models beat machine learning

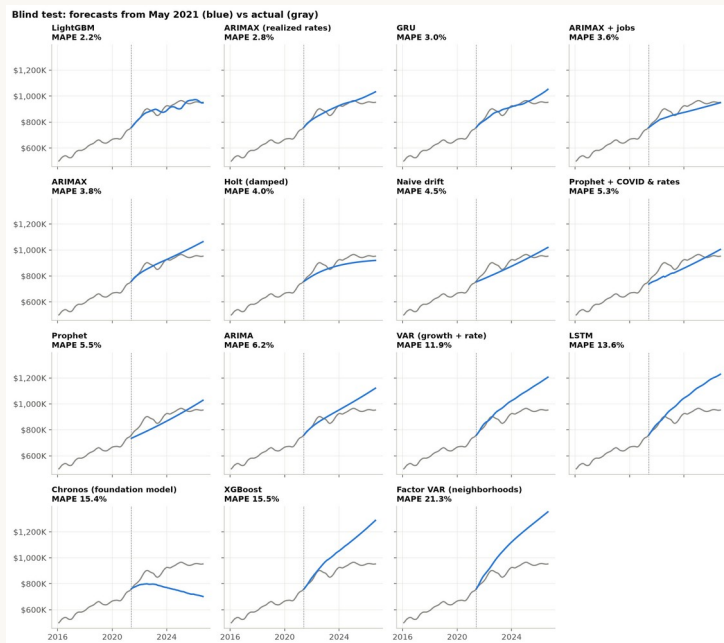


LOWEST ERROR

2.67%

ARIMAX, average error 1 to 12 months ahead. ARIMA 2.77%. LSTM 3.02%, XGBoost 3.65%, Prophet 6.89%.

One test window crowned LightGBM; 18 windows ranked it tenth

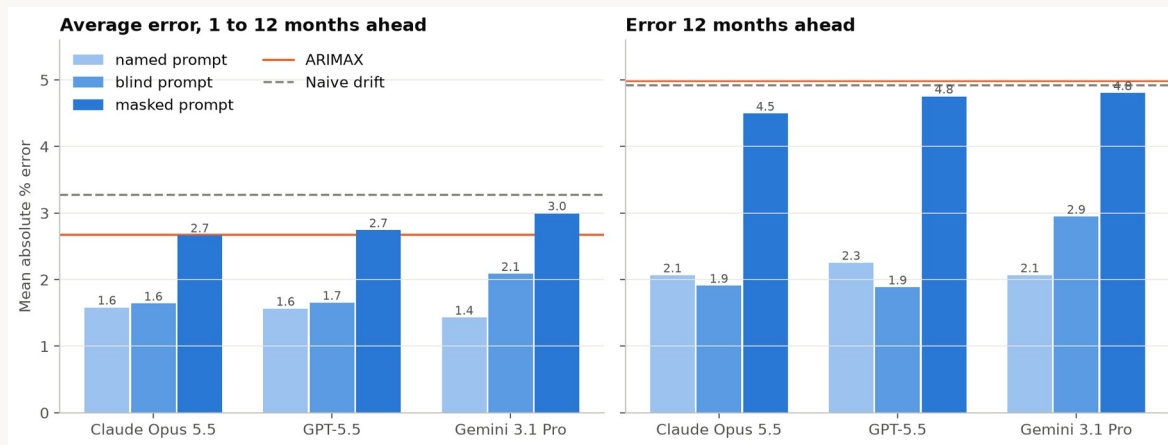


SAME MODEL, TWO TESTS

1st → 10th

LightGBM won the single 64-month test (2.2% error) but ranked 10th across 18 rolling tests: it got lucky in the one window that counted.

Chat LLMs beat every model, until we hid which years they were forecasting

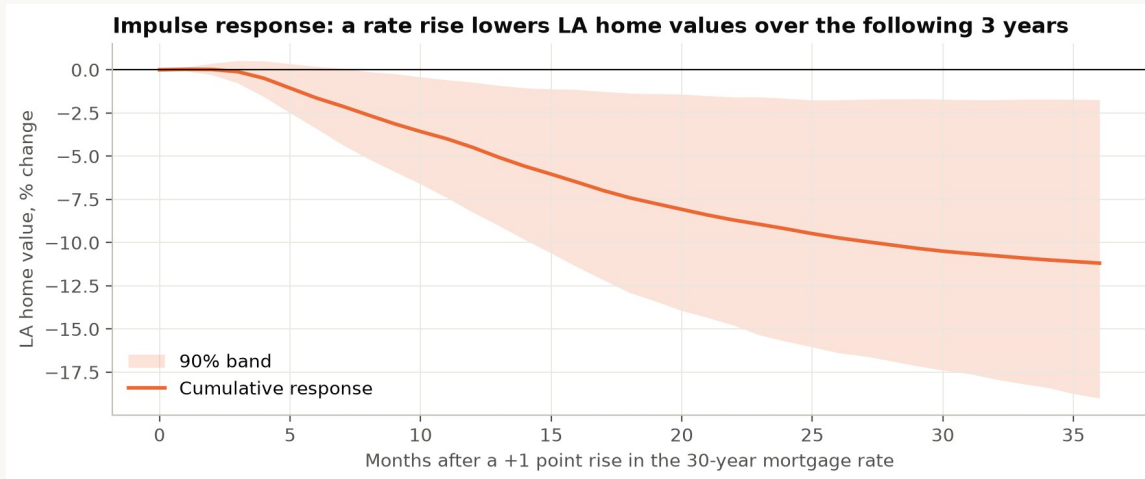


SAME MODEL, DATES HIDDEN

1.6 → 2.7%

Claude's error once it could not tell the years (ARIMAX 2.67%). All test dates are inside the LLMs' training data.

A 1-point rate rise means LA values about 9% lower two years later

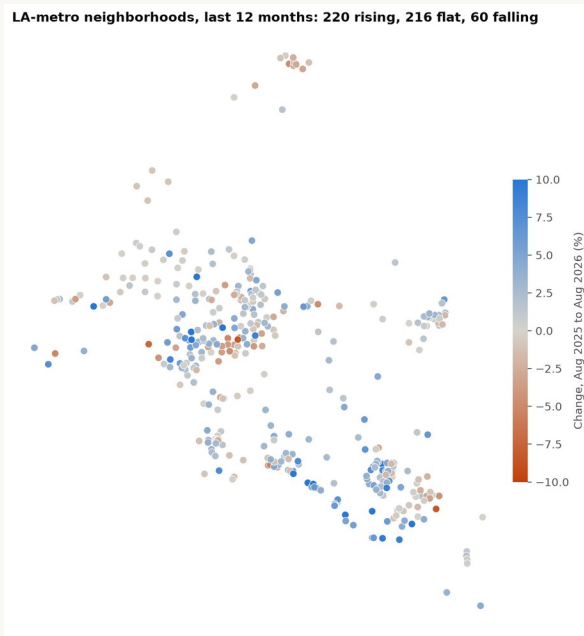


RATES WORK SLOWLY

-9.2%

after 24 months (-4.5% after 12). Rates help predict prices, not the reverse (Granger $p = 0.05$ vs 0.51). Local jobs add nothing once 2020–21 is set aside.

The metro average hides a split market, neighborhood by neighborhood



220 / 216 / 60

neighborhoods rising / flat / falling more than 2%,
last 12 months

152 of 496

still below their 2022 peak, mostly in LA County

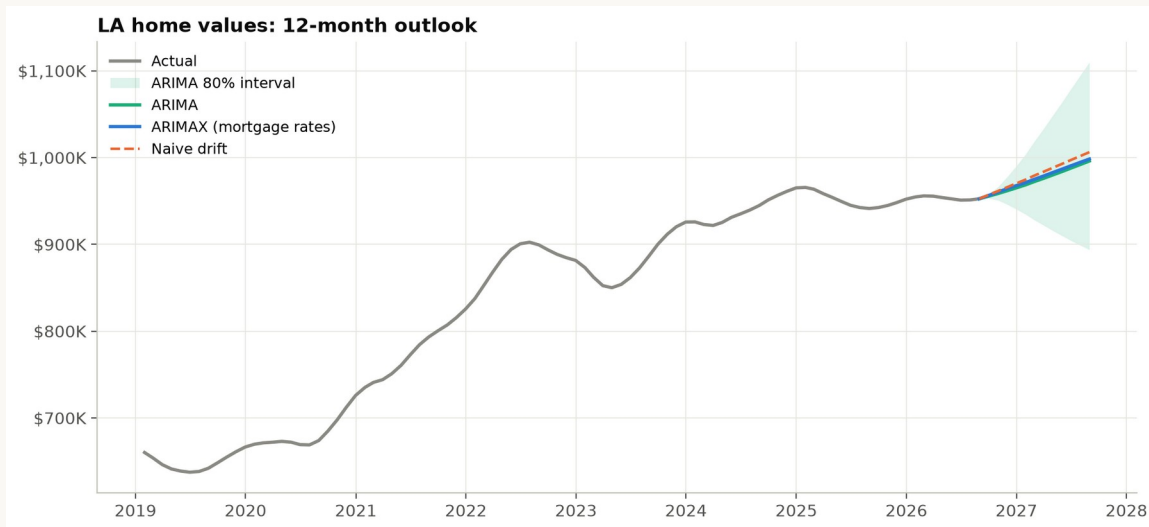
-1.2 pts

deeper 2022 drop per doubling of price in LA
County (reverses in Orange County)

Live map: neiltd.github.io/la-housing-forecast

OUTLOOK

Our forecast: about 5% higher by August 2027, with a wide range



\$998,391

ARIMAX forecast, Aug 2027 (+4.8%)

-6% to +17%

80% range for the next 12 months

+4.8% / +6.5%

median neighborhood forecast, LA County
/ Orange County

What we learned

01

Simple beats complex for a smooth monthly index

ARIMA-family models beat LSTM, GRU, boosted trees and the Chronos foundation model.

02

How you test decides who wins

One blind window ranked LightGBM 1st; 18 rolling windows, 10th.

03

An LLM can't be backtested on years it has read about

Claude, GPT and Gemini beat every model until we hid the dates; then they matched ARIMAX at best.

04

Mortgage rates matter, slowly; local jobs don't add

A 1-point rise lowers values about 9% over two years.

05

The metro average hides a split market

Separate models per neighborhood beat one shared factor (2.8% vs 4.1% error).